

Ali Haj

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EDUCATION

Temple University — Philadelphia, PA

Bachelor of Science in Electrical Engineering

Expected Graduation: May 2029

Academic Programs & Honors

- Temple University Honors Program
 - Sustainable Temple Energy and Power Scholars (STEPS) Program
 - Dean's List — Fall 2025
 - Dean's List — Spring 2026
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ENGINEERING EXPERIENCE

Undergraduate Robotics Assistant

Temple University | Summer 2026 – Present

Selected through Temple University's **Sustainable Temple Energy and Power Scholars (STEPS) Program** to work with engineering faculty and fellow engineering students on robotics development projects. Contribute to the design, programming, testing, and continuous improvement of autonomous robotic systems while gaining hands-on engineering experience.

Responsibilities

- Develop, test, and debug software for robotics applications using Python.
- Design and refine autonomous robot behaviors through algorithm development and iterative testing.
- Integrate sensor feedback into robotic control systems to improve navigation, decision-making, and overall performance.

- Collaborate daily with a Temple University professor and multidisciplinary engineering students on robotics design, programming, and system optimization.
 - Apply engineering design principles to solve technical challenges involving robotics, autonomous systems, and software development.
 - Participate in system testing, debugging, performance evaluation, and design improvements for engineering projects and competitive robotics applications.
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ENGINEERING PROJECTS

Autonomous Zumo Sumo Robot

- Designed and programmed an autonomous competition robot capable of detecting opponents, identifying arena boundaries, and executing strategic attack and recovery behaviors.
- Improved robot performance through algorithm development, sensor calibration, iterative testing, and software optimization.
- Applied robotics, programming, and engineering design principles to enhance autonomous decision-making and overall system performance.

Wind Turbine Energy Analysis Using Python

- Analyzed meteorological datasets using Python to estimate wind turbine energy production.
- Processed environmental data to evaluate renewable energy generation under varying weather conditions.
- Applied programming and data analysis techniques to solve engineering problems involving renewable energy systems.

Micro:bit Engineering Incubator Project

- Designed and programmed engineering projects using the BBC Micro:bit platform.
 - Integrated sensors and input/output devices to create interactive embedded applications.
 - Applied engineering design principles through prototyping, testing, and iterative development.
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TECHNICAL SKILLS

Programming

- Python

Engineering

- Robotics
- Engineering Design
- Autonomous Systems
- Sensor Integration
- BBC Micro:bit

Software

- Microsoft Excel
- Git
- GitHub

Professional Skills

- Data Analysis
 - Technical Research
 - Problem Solving
 - Critical Thinking
 - Team Collaboration
 - Time Management
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